



**Sanjivani Rural Education Society's**  
**Sanjivani College of Engineering, Kopargaon-423 603**  
*(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)*  
**NAAC 'A' Grade Accredited, ISO 9001:2015 Certified**

## **Department of Computer Engineering**

*(NBA Accredited)*

**Subject- Object Oriented Programming (CO212)**  
**Unit 1 – Fundamentals of OOP**  
**Topic –1.8 Strings in C++**

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# What is a String?

A string is a collection of characters . It is usually a meaningful sequence representing the name of an entity.

Generally it is combination of two or more characters enclosed in double quotes.

“Good Morning” // string with 2 words

“Mehak” // string with one word

”B.P” // string with letters and symbols

“” // an empty string



# How to store and process strings in C++?

- C++ does not provide with a special data type to store strings.
- Thus we use arrays of the type char to store strings
- Strings in C++ are always terminated using a null character represented by „\0“
- String literals are the values stored in the character array



# String/Character Array Declaration

To declare a character array/string, the syntax is :

```
char arrayname[size];
```

Example :

```
char name[20];
```

Here name is a character array or string capable of storing maximum of 19 characters. One character is reserved for storing „\0"/NULL Character

Thus the number of elements that can be stored in a string is always  $n-1$ , if the size of the array specified is  $n$ .



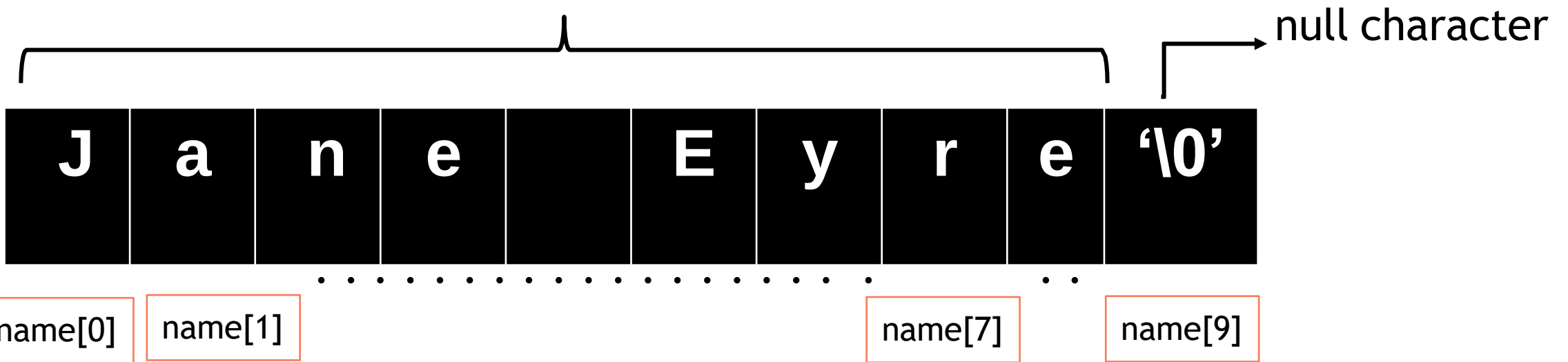
# Examples of char array declarations

| Array declaration                  | Used to store       |
|------------------------------------|---------------------|
| <code>char city[20];</code>        | name of a city      |
| <code>char address[40];</code>     | address of a person |
| <code>char designation[15];</code> | designation         |
| <code>char author[25];</code>      | name of author      |



# String : Initialization

```
char name[20]="Jane Eyre";  
           string
```



The above string will have 9 characters and 1 location for the null. Thus size of name will be 10.



# String: Initialization

We can also initialize a character array in this way:

```
char name[ ]="Jane Eyre";
```

- In this case the array is initialized to the mentioned string and the size of the array is the number of characters in the string literal. Here it is 10.
- In the above declarations the null character is automatically inserted at the end of the string.

# String : Initialization

We can also initialize a string in this way:

```
char name[ ]={„J“, „a“, „n“, „e“, „ „, „E“, „y“, „r“, „\0“}
```

Note :

- Here the character array or string is initialized character by character.
- The „\0“ has to be inserted at the end. In this method the null character has to be inserted by the programmer only.





# String Input and Output

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- We can input a string in two ways:
  1. Using **cin>>**
  2. Using functions i. **gets()**    ii. **getline()**
- A string is displayed using a simple **cout<<** statement
- Note that we do not need to use a loop to input or display a string.



# Input using cin>>

## Program

```
#include<iostream>
int main()
{ char city[13];
  cout<<“\n Enter name of the city”;
  cin>>city;
  cout<<“\n you entered :”;
  cout<<city;
  return 0;
}
```

## Output

```
Enter name of the city
Pune City
you entered :Pune
```

Contents of array

|   |   |   |   |    |  |  |  |  |  |  |  |  |
|---|---|---|---|----|--|--|--|--|--|--|--|--|
| P | u | n | e | \0 |  |  |  |  |  |  |  |  |
|---|---|---|---|----|--|--|--|--|--|--|--|--|

## Explanation

In the above code cin stops input as soon as it encounters a space after Pune. So the string variable city will contain only Pune. Note: „\0”will be inserted after Pune.



# Input using gets()

- The gets() function can be used to input a single line of text. As soon as the enter is pressed the gets() function stops input and the string is terminated.

## Program

```
#include<iostream>
int main()
{ char city[13];
  cout<<"\n Enter name of the city";
  gets(city);
  cout<<"\n you entered :";
  cout<<city;
  return 0;
}
```

## Output

```
Enter name of the city
Pune city
you entered :Pune city
```

Contents of array

|   |   |   |   |  |   |   |   |   |    |  |  |  |
|---|---|---|---|--|---|---|---|---|----|--|--|--|
| P | u | n | e |  | c | i | t | y | \0 |  |  |  |
|---|---|---|---|--|---|---|---|---|----|--|--|--|

## Explanation

In the above code, gets() will input the entire string.



# Input using getline()

- The getline() function can be used to input multiple lines of text. The syntax is :

`cin.getline(string, MAX, Delimiter character)`

where

- string is the variable in which the input is done
- Max is the maximum number of characters allowed to input.
- Delimeter character is the character which when encountered in the input stream stops the input. The default delimeter character is "\n"



# Input using getline()

The getline function inputs the string until either the maximum number of characters are input or it encounters the delimiter character whichever comes first.

## Program

```
#include<iostream>
int main()
{ char para[80];
  cout<<"\n Enter in your own words:";
  cin.getline(para,80, „#“);
  cout<<"\n you entered  :";
  cout<<para;
  return 0;
}
```

## Output

```
Enter in your own words : Pollution is a great factor
in determining the health
of today's youth#
you entered  : Pollution is a great factor
in determining the health
of today's youth
```

## Explanation

In the above code, cin.getline() continues to input the string till it encountered a „#“. This is because here „#“ is the delimiter character.



# Using getline()

## Program

```
#include<iostream>
int main()
{ char para[80];
  cout<<"\n Enter in your own words:";
  cin.getline(para,60, „#“);
  cout<<"\n you entered  :";
  cout<<para;
  return 0;
}
```

## Output

```
Enter in your own words : Pollution is a great factor
in determining the health
of today's youth#
you entered  : Pollution is a great factor
in determining the health
of toda
```

## Explanation

In the above code, since the maximum specified length is 60, so the string para will contain input till toda. Note that last space will be occupied by the „\0“ character.



Thank You..